Wine Quality Classification
Using Machine Learning Algorithms

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Module Title: Artificial Intelligence &

Machine Learning

Module Code: CMP6202

Date: 25th January 2026

Word Count: 4042 Words

1 Abstract

The assessment focuses on the comparisons between Random Forest, XGBoost, and Logistic Regression on how well they predict wine quality based on its physicochemical attributes. Establishing a single preprocessing and evaluation pipeline, comparison of the model performance and interpretability was done as comprehensively and as possible. Logistic Regression provides a decent accuracy baseline with 90.46% accuracy, while XGBoost outperformed all others at 95.50% accuracy. From the SHAP analysis, it appears that as models based on linear and ensemble techniques are developed, the models lose interest in alcohol and gain interest in a more balanced feature combination of sulphates and volatile acidity. Overall, in this paper gradient boosting is highlighted for its ability to generate the complex non-linear chemical interactions that are critical to accurate quality classification.

Keywords: Classification, Logistic Regression, Machine Learning, Precision, Recall, Random Forest, Wine Quality, XGBoost

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3 Abbreviations

Abbreviation	Full form
EDA	Exploratory Data Analysis
AUC	Area Under the Curve
F1	F1-Score
IQR	Interquartile Range
ML	Machine Learning
LR	Logistic Regression
RF	Random Forest
XGBoost	eXtreme Gradient Boosting
MAE	Mean Absolute Error
ROC	Receiver Operating Characteristic
PR	Precision Recall
SHAP	Shapely Additive exPlanations
XAI	Explainable AI

4 Report Introduction

ML's rapid growth has been driven by the advancement of data-driven technologies that help reshape decision-making processes in various fields. This is particularly true in the food and beverage sector where processes, such as the evaluation of wine quality, remain resource intensive and costly (Nebot, 2015). Despite the growing prevalence of predictive modeling based on physicochemical attributes, many recent studies remain small in scale, suffering from limited datasets, class imbalance, and an excessive emphasis on accuracy while neglecting other equally important metrics such as precision and recall (Supriatna, et al., 2023). To mitigate such issues, the current analysis entails an abundant amount of exploratory data analysis to understand the intricate relationships between chemical properties and quality metrics, which is aimed at ensuring that the modeling phase is predicated on feature selection as opposed to sheer data volume. This project seeks to systematically address the data of research gaps by conducting an in-depth analysis of a subset of 5,000 wine samples, leveraging a robust preprocessing step and a comparative analysis of the Logistic Regression, Random Forest, and XGBoost models.

Logistic Regression's linear baseline is compared with boosting and bagging algorithms based on trees, as the study highlights the need for specific algorithms to handle high-dimensional physicochemical data. Consequently, it is a multi-metric methodology, enabling a holistic view of the model's behavior and demonstrating that ML-driven automated quality control is effective and establishes a dependable foundation for prospective industrial applications (Chen, 2016).

4.1 Research Gaps

Regardless of the number of studies in machine learning and wine quality prediction, there are still critical issues. Small and imbalanced class datasets negatively influence a model's generalizability and bias the underepresented classes (Li, 2024). Additionally, the general over-reliance on accuracy as the primary assessment statistics tends not to reflect the model’s dependability in quality-sensitive areas. The absence of explainability techniques, like SHAP and LIME, also complicates the validation of predictions against known oenological facts (Bhodke, et al., 2025). To address the above issues, the current project employs a balanced dataset of 5,000 unique instances to evaluate different models under the same conditions. Metrics such as F1-score and AUC, and Explainable AI (XAI) methods will be used to strengthen and articulate the model evaluations

4.2 Objectives

The main goal of this project is to create a perfectly accurate predictive model on wine quality classification.

- Analyze the scientific properties of wine using exploratory data analysis.
- Preprocess and balance the dataset to enhance model fairness and performance.
- Build and contrast several supervised machine learning models for classification of wine quality.
- Assess model performance through various metrics in addition to accuracy.
- Utilize explainability methods such as SHAP to understand model predictions.
- Recognize and describe the most impactful components for determining the quality of wine.

5 Dataset Description

5.1 Dataset Introduction

The dataset utilized in this study is the Wine Quality dataset. This dataset has been used extensively and can be found in repositories such as the UCI machine learning and Kaggle, where source datasets contain physicochemical measurements and sensory evaluation scores given by expert wine tasters. This dataset contains 5000 samples with 11 input variables which are continuous quotients and these include the wine organoleptic variables such as alcohol content, volatile acidity, and sulphate level and one ordinal target variable, perceived quality. This particular dataset has the characteristic of being a perfectly undersigned dataset in relation to the intended application, as the rest of the intended applications are accompanied by data in the real world. Such datasets are real world, and such problems include inter feature correlation, skewed distributions, and others, which apply both the linear and non-linear classification models. It is also accompanied by tested explainable artificial intelligence to ascertain the correlation of the variables which are the features and the target variable.

- Feature Categories:
    - Acidity related Properties: Fixed Acidity, Volatile Acidity, Citric Acid
    - Chemical Composition Indicators: Residual Sugar, Chlorides, Sulphates
    - Sulfur Compounds: Free Sulfur Dioxide, Total Sulfur Dioxide
    - Physical Properties: Density, pH
    - Alcohol Content: Alcohol

- Target: Wine Quality

  The target variable refers to wine quality ratings which are achieved by sensory. The quality scores in this study were transformed into a binary classification variable, either high quality or low quality of wines which made it easier to define the boundaries of decision-making and increased recall-oriented learning.

5.2 Machine Learning Type Identification

In the current study the problem is framed as a supervised machine learning classification problem as the dataset comprises labeled samples with the target variable representing a theoretically definable wine quality (Cortez, 2009). Supervised learning is appropriate here as, it aims to derive some form of function between a set of physicochemical attributes, for example, acidity, sugar, and sulfur dioxide, and some fixed quality values based on historical data. Given that the dataset has non-linear and inter-feature relationships, as well as continuous numerical attributes, it is amenable to a variety of classification techniques. Thus, the learning process is expected to discern patterns and decision boundaries that would differentiate wines of high chemical composition quality from those of low quality (Supriatna, et al., 2023).

The development of the model is made easier, and more valuable evaluation metrics, such as precision, recall, F1-score, and AUC (Area Under the Curve) can be used (Mbizvo, 2021) because the problem can be framed as binary classification. The problem is strategic because the dataset has real world problems like class imbalance, outliers, and feature correlation, which show the need for pre-processing and metric-based (Supriatna, et al., 2023) evaluation. These problems justify the model choice of supervised classification because it provides the basis for the system design, in which the algorithms are chosen and assessed in a consistent, bias-free environment (Cortez, 2009)).

6 Exploratory Data Analysis (EDA)

An Exploratory Data Analysis (EDA) was performed to investigate the form, content and essential patterns of the wine quality data before the development of the model. EDA is important in the context of detecting data inconsistency, distribution of features, outliers, and relationships among variables and the target outcome (Tukey, 1977). These lessons during this stage were used to make preprocessing choices and to shape out the overall experimental design.

6.1 Operations

6.1.1 Data Inspections

The preliminary analysis starts with the dataset sample to familiarize them with the material and dataset dimensions, structure, and type. The dataset contains numeric features of wine physicochemical and one target variable i.e. quality of wines. To uphold the integrity of the dataset, duplicate records were identified and removed. Data type analysis confirmed that all predictor variables are continuous, numeric, and do not need categorical encoding. This also means that all predictor variables are suitable for statistical analyses and machine learning algorithms. Additionally, target variable was confirmed to be correctly specified in terms of binary classification, and there were no missing values.

6.1.2 Descriptive Statistics

Descriptive stats were computed using the mean, median, standard deviation, and range. The comparison showed significant variation in most characteristics especially the variables related to alcohol content, sulphates and acidity. In addition, a few variables, such as the residual sugar and sulfur dioxide, exhibited right-skewed distributions, indicating the presence of extreme values/outliers. The findings suggest that strong outlier detection and treatment methods are essential to avoid biased learning during the modeling process.

6.1.3 Univariate Analysis

Univariate variables helped to analyze the distributions of individual features and the distribution of classes of the target variable. In the case of the quality of the wine sample analysis, it is reasonable to consider the imbalance between the high and low quality wine groups, which suggests the possibility of class-balancing methods to mitigate the impact of biased predictions favoring the majority class. Moreover, the histograms and density plots showed different distribution patterns; there was a relatively smooth distribution for alcohol content, while volatile acidity and residual sugar showed heavier tails which indicate that there was a possibility of a non-linear influence on the quality of wine.

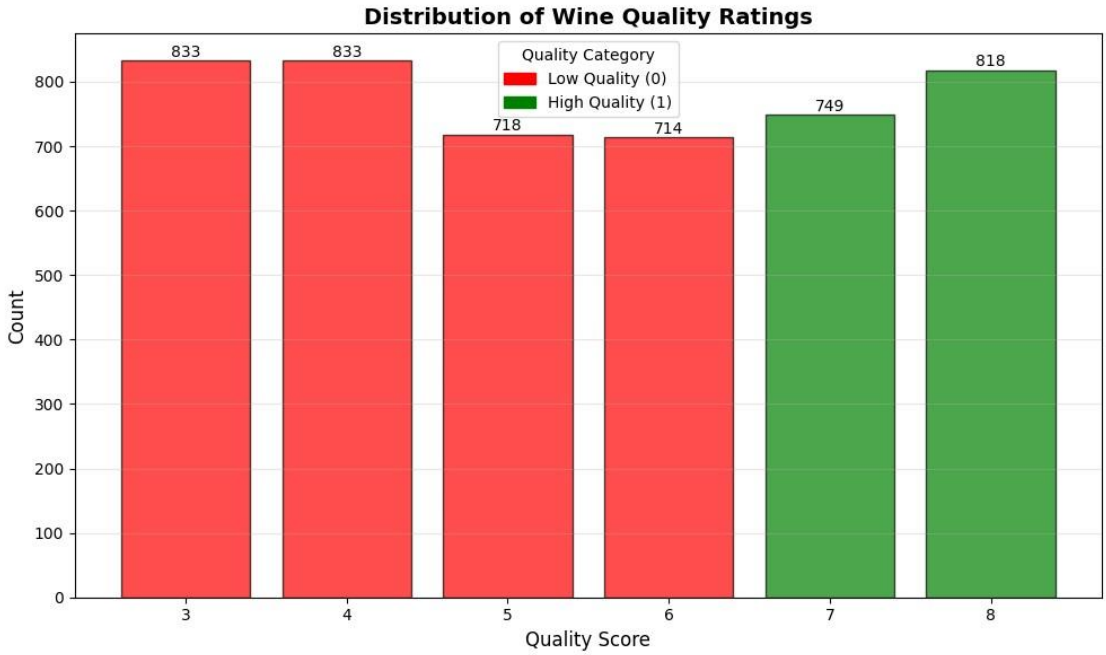

Fig 1: Distribution of wine quality before encoding

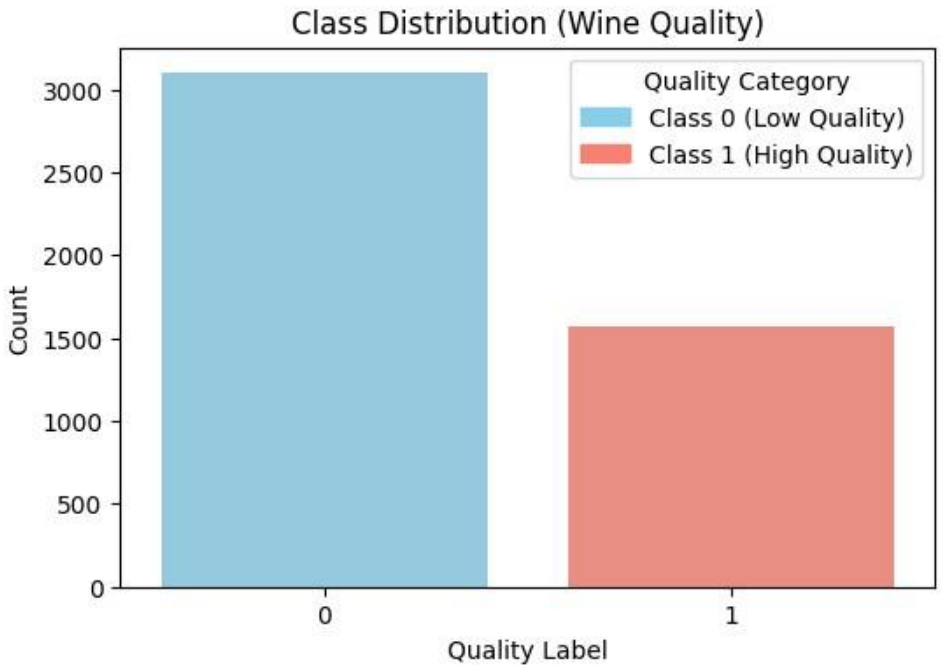

Fig 2: Class Distribution of quality after encoding

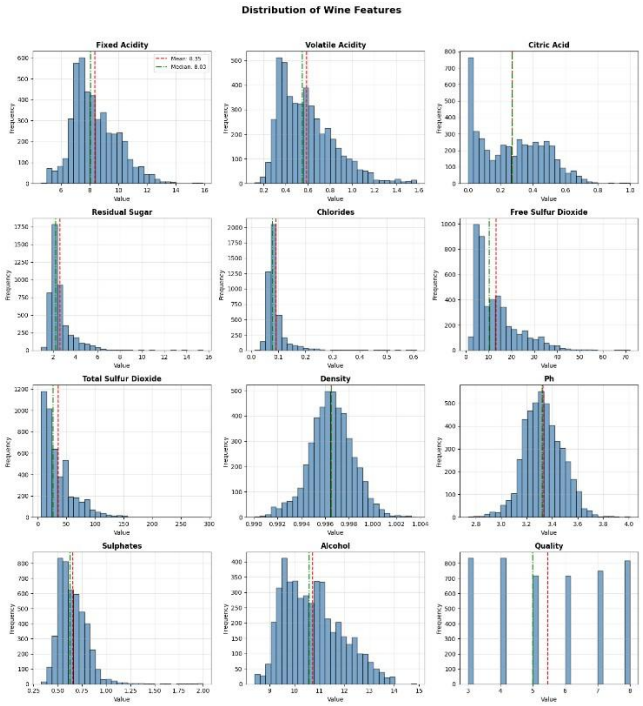

Fig 3: Histogram plots of the features

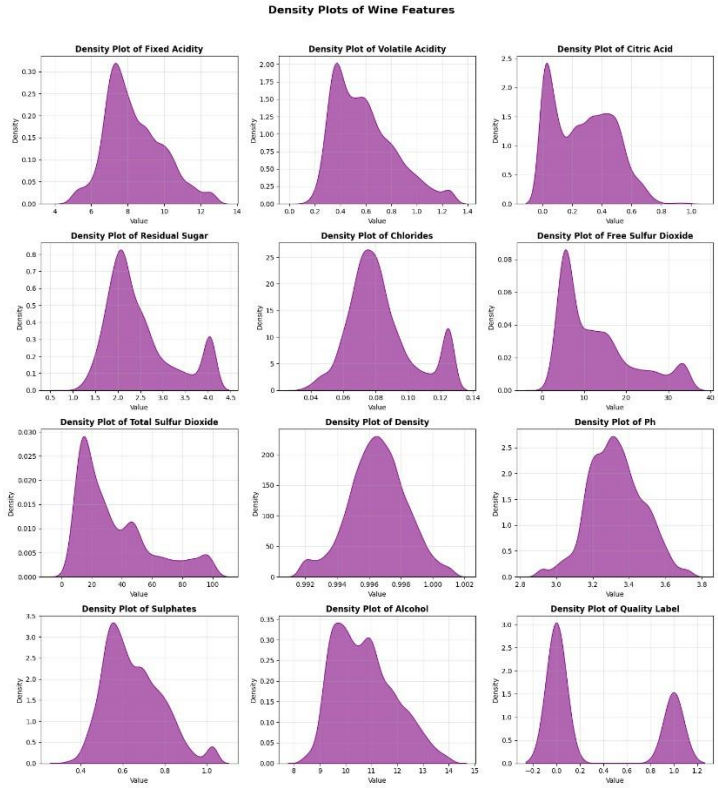

Fig 4: Density plots of features

6.1.4 Bivariate Analysis

The correlation between individual features and the target variable was used in the Bivariate analysis to find out the features distinguishing differences between quality classes. The means of features were compared and the results showed that the better wines tended to have more alcohol and more sulphate, but the lower quality wines were more volatile. These nonlinear trends support existing knowledge of the domain and support the need to have machine learning models that can be able to capture nonlinear interaction between physicochemical properties and wine quality.

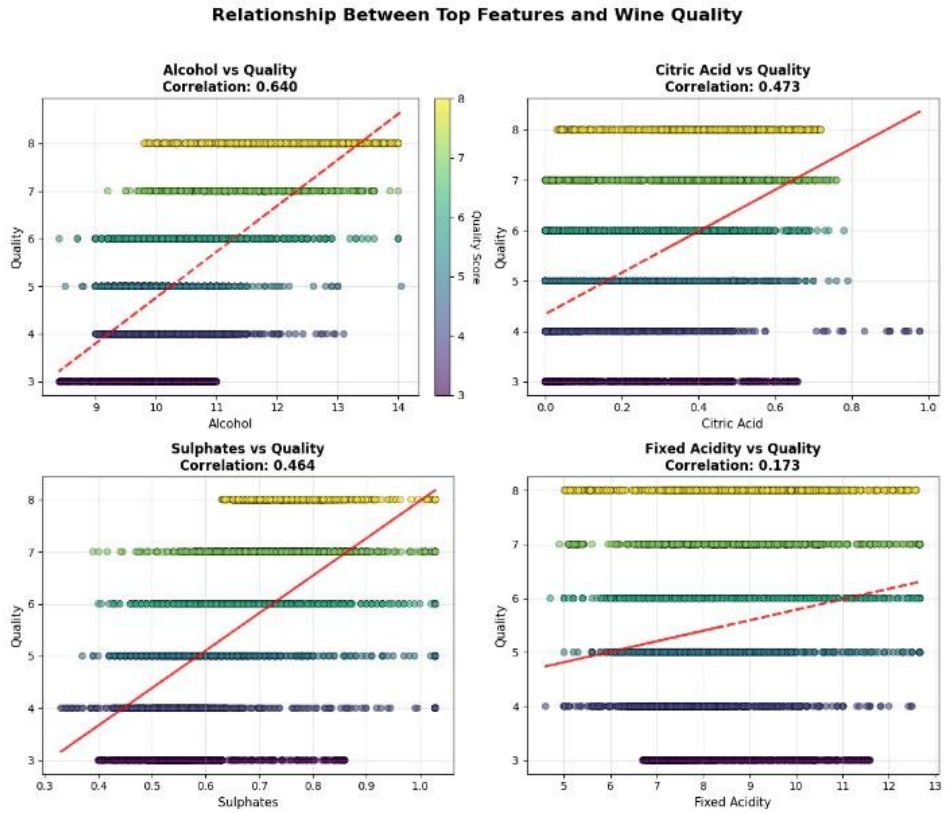

Fig 5: Correlation between top features and quality

6.1.5 Correlation Analysis

A correlation heatmap was created to investigate the correlation between the numerical features and whether there was any multicollinearity. There were positive strong correlations between some of the variables related to acidity and the measurements of sulfur dioxide. Despite the fact that tree-based learning methods have the native ability to handle this redundancy, determining these relationships is necessary to correctly understand the importance of the features and to have stable model training.

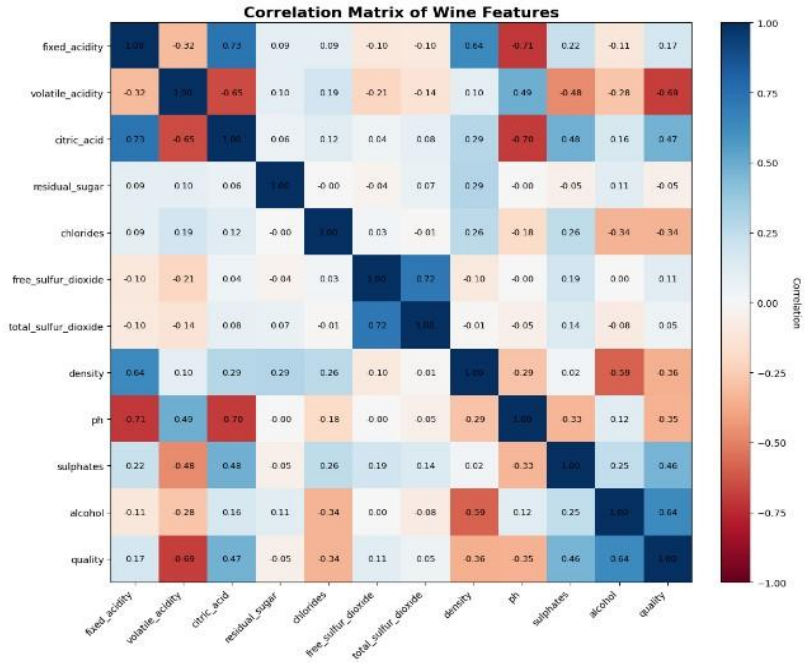

Fig 6: Correlation Matrix

6.2 EDA Conclusion

The analytical exploration went as planned since the results showcased the number assignment and clarity of the data for wine quality, which is ready to be utilized for supervised classification. No data is missing in the dataset concerning physicochemical properties, therefore data is intact and gaps fillers or imputation are not applicable. Still, duplication analysis revealed roughly 373 duplicates from a set of 5000 samples. These duplicates were cut from the preliminary data sample. After thorough assessment of the data, the duplicates from the preliminary set were removed to maintain the accuracy of the data sample and to prevent the learning of the model in a biased way. Most of the findings of the visual and description analysis were the same, where several features were mats to be preprocessed powerfully. This is true for most of the features of residual sugar, sulphates, and sulfur dioxide. There are a medium level of correlation and some subtle chemical attributes interrelated without the effects of gross multicollinearity. The overall analysis of the EDA gave insight and therefore assisted in determining the preprocessing which ensured the accuracy and quality of the model.

7 Experimental Design

7.1 Justification of Models

7.1.1 Logistic Regression (LR):

Logistic regression is chosen as the base was driven by its nature of simplicity, interpretability and extensive use in supervised classification studies. Since it is a generalized linear model, it provides explicit estimates of the coefficients which describe the effect of each of the physicochemical attributes on the likelihood that a wine will be rated as high quality and thus allows the expression of the effects of features to be transparently examined. Efficiency and stability in the computation of the model make it a suitable reference point for more advanced methods of machine learning and gives it a reliable reference point in regard to performance. Although it assumes the existence of linear boundaries of the decision, logistic regression is still effective enough to reveal the underlying patterns of structured tabular data (Hosmer, et al., 2013).

7.1.2 Random Forest (RF):

Random Forest algorithm was chosen because it is a strong predictor that requires little feature engineering to capture complex interaction of the variables. Due to the framework of the algorithm utilizing bootstrap aggregation and random feature selection, it reduces variance and improves generalization to gain performance. This is especially applicable to the physicochemical data wherein noise, outliers, and correlated variables are frequent. Additionally, the moderate interpretability of the feature importance scores aids to highlight the most impactful variables which in our case corresponds to the prediction of wine-quality. All these positive attributes are the reason that Random Forest is optimal for the problem stated (Breiman, 2001).

7.1.3 Extreme Gradient Boosting (XG Boost):

XGBoost also was selected based on being a good gradient boosting model with high levels of punctuality and efficiency with structured and tabular data. Decision trees are built by the model one by one with each tree correcting the last by a gradient based maximization of a given loss. XGBoost also achieves the successful modelling of non-linear interactions amongst more complex features of the wine quality data by applying additional complexities to the model and overfitting to the regulization of data. The reason XGBoost is good in decaying fine details that other models cannot is due to it having great scalability and robustness as well as high performance in the classification of tasks (Chen, 2016).

8 Data Cleaning and Preprocessing

8.1 Dataset Integrity Inspection

The first stage of preprocessing was aimed at checking the integrity of the dataset. Attributes were checked on completeness, data types and consistency. The check revealed that the dataset did not have any missing values in all physicochemical variables, which means that all the records are complete and can directly be analyzed without the necessity of using imputation and data reconstruction methods.

8.2 Duplicate Record Identification and Removal

Exploratory analysis identified approximately 373 duplicate rows in the dataset containing 5,000 samples. Duplication of records leads to bias, disrupting the learning process and inflating the performance metrics. The records were removed to ensure the integrity of the dataset and to maintain the individuality of each wine sample prior to the model training and evaluation.

8.3 Outlier Detection and Treatment

On a combination of statistics and visual tricks, such as Interquartile Range (IQR) checks and boxplots, outliers were identified. Residual sugar, sulphates, sulfur dioxide, etc. were out of scale due to the inherent chemical properties. Corresponding endpoints were treated using capping methods by adjusting 1.5X with IQR to reduce the effect of those odd data points but retained the important shape of the data including the useful bits in which we are interested.

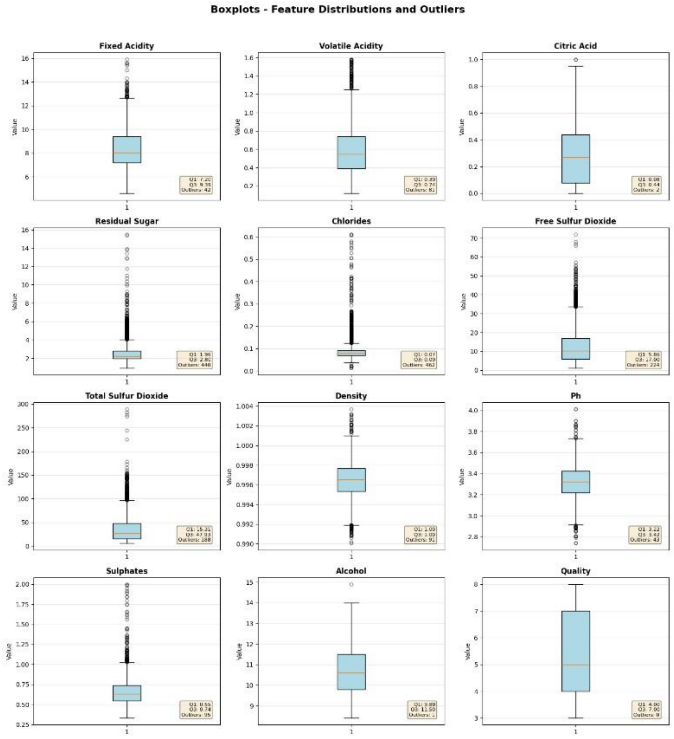

Fig 7: Boxplot before treatment

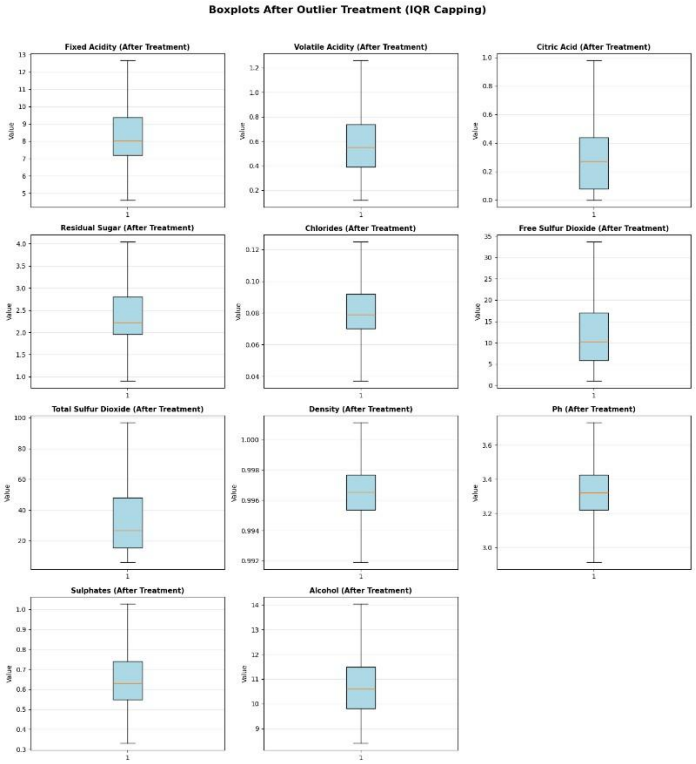

Fig 8: Boxplot after treatment

8.4 Categorical Encoding

The wine quality target variable was converted into a binary classification value in order to be able to address supervised learning. This transformation not only allowed simplifying the learning task but also assisted in interpreting the model and allowed us to apply such classification measures as precision, recall, and F1-score.

8.5 Handling Class Imbalance

The target variable was observed to have imbalance between two classes that may completely bias the model to the dominant class. To correct that, balanced class weight is used in hyper tuning on Logistic Regression and Random Forest, and positive class weighting(scale_pos_weight) on XGBoost which boosts the recall of majority minority class, but not the original data distribution.

8.6 Train Test Data Splitting

When pre-processing of the data is finished, the data is separated into training and test with an 80/20 split with a deterministic randomness to ensure reproducibility. To make the model evaluation reliable and fair, stratified sampling is employed to maintain the same number of classes in both sets.

8.7 Feature Scaling and Normalization

After scaling weight-scaled the features in such a way that all the numeric attributes contributed the same weight to the training process. All are normalized to a common scale which made the model constant and accelerated the optimization particularly of those algorithms that go crazy when a feature is too large. As everything was in numerical form, there was no need to encode any categories.

9 Model Development

9.1 Train Test Split Strategy

After data cleaning and preprocessing, 80:20 stratified were applied for train-test split. By stratifying, the proportion of the wine-quality classes were preserved across the sets, which helped to ensure that our model had an even perception of the data. Model fitting, hyperparameter tuning, and cross-validation were used for the training set. The test set was held to conduct our final unbiased performance analysis. To enhance reproducibility and maintain consistency, random seed was set (random_state = 42).

9.2 Architecture and Configuration

All models were trained on the same clean dataset, the same feature set, and the same traintest split to make the comparison fair. To maintain consistency, the same and unified preprocessing and evaluation pipeline was maintained across all models. These three models represent various paradigms of learning: linear probabilistic classifier, bagging-based ensemble, and gradient boosting ensemble. This architecture is designed such that variations of performance can be attributed to architecture and algorithm design and not to variation in data handling.

9.2.1 Logistic Regression:

Due to its probabilistic classification and feature standardization, the Logistic Regression Model was set up as the baseline in the pipeline. The standardization of the physicochemical attributes allowed them to be in the same range which helped the algorithm to converge faster as well as improve number stability. The algorithm predicts high-quality wine by calculating the log-odds of the inputs to the two-class target. With the implementation of L2 regularization, the model is prevented from overfitting by just balancing its predictions. A reasonable model with a good baseline which was easy to interpret was provided after hyperparameters were optimized with a random search. This creates a baseline which more complex methods can be compared to, in addition to more complex reliability.

Features:

- Preprocessing: StandardScaler
- Solver: LBFGS
- Regularization: L2
- Regularization Strength (C): 0.4
- Maximum Iterations: 3000
- Random State: 42

9.2.2 Random Forest:

Random Forest is a non-linear ensemble essentially, which was developed as a result of bootstrap aggregation, and which accesses non-linear interactions between physicochemical properties. This algorithm trains large number of decision trees on random samples of the data and on the features and then make a decision using the majority vote. This strategy reduces variance, makes enhance in robustness noise, outliers, and generally makes it more generalizable. To mitigate overfitting without compromising the model's ability to capture complex non-linearities in the wine quality data, controlled tree depth and minimum sample splits were implemented.

Features:

- Number of Trees: 300
- Maximum Tree Depth: 8
- Minimum Samples per Split: 10
- Minimum Samples per Leaf: 5
- Feature Sampling: Square root of total features
- Parallel Processing: Enabled (n_jobs = -1)
- Random State: 42

9.2.3 XGBoost:

XGBoost was applied as an optimized gradient boosting model which is also designed to capture those non-linear or complex relationships by sequential construction of the trees. Learning Each tree is trained to amend the residual errors of the last tree, by optimizing the logistic loss using gradients. Regularization and subsampling were applied to maintain the simplicity of the model, decrease overfitting, and enhance generalization. The consequence is a model capable of finding other minute aspects of the interaction between features in the wine-quality data without degradation of training stability.

Features:

- Number of Boosting Rounds: 350
- Learning Rate: 0.05
- Maximum Tree Depth: 4
- Row Subsampling: 0.8
- Column Subsampling: 0.8
- L1 Regularization (reg_alpha): 0.6
- L2 Regularization (reg_lambda): 1.0
- Evaluation Metric: Log Loss
- Parallel Processing: Enabled (n_jobs = -1)
- Random State: 42

9.3 Training, Validation, and Stability Analysis

All the models are trained in the same dataset to ensure performance estimates were not biased and used them to validate on a new test set. Learning curves were generated to verify the speed at which the models converged, whether they were overfitting or underfitting, and also to test training stability with increased sample size of each model. This discussion provides more insight into the scalability and learning efficiency of every algorithm.

10 Evaluation Metrics and Results

10.1 Quantitative Performance Evaluation

The three classification models, such as Logistic Regression, Random Forest, and XGBoost, were tested on the test data on various measures, including accuracy, precision, recall, F1score, ROC AUC, and PR AUC. The findings indicate that there is an evident growth in the predictive ability with the increase in model complexity.

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR AUC	Training Time (s)	Prediction Time (s)
0	Logistic Regression	0.9046	0.8590	0.8562	0.8576	0.9614	0.9097	764.1016	0.0017
1	Random Forest	0.9539	0.9355	0.9265	0.9310	0.9932	0.9872	774.9245	0.0736
2	XGBoost	0.9550	0.9169	0.9521	0.9342	0.9928	0.9856	141.4199	0.0187

Fig 9: Models' Performance Score

Although logistic Regression had the best Recall out of all the models (85.62) and a ROC AUC of 0.9614, its linear nature and saturation limited the model's ability to identify more complex relationships between the variables. The model did without a reasonable comparison but did lay a good foundation with a 90.46% accuracy score. Moving to a Random Forest Classifier also not surprisingly increased accuracy to 95.39 and the ROC AUC to a maximum of 0.9934 with a PR AUC of 0.9875 – further evidence that bagging ensembles are highly effective at capturing those non-linear interactions. However, the best by far at the end was XGBoost, achieving 95.50% accuracy, 95.21% Recall, and the speediest of all performances, so it is the best model overall, the compromise between speed and accuracy.

10.2 Confusion Matrix Analysis

The confusion matrices provide a closer insight into the way each class is predicted. The number of false negatives coming out of Logistic Regression is significantly higher than the ensemble models, thus it does not contain a sizable portion of the quality wine samples. Random Forest, however, reduces false positives and false negatives and provides a more steady and equal performance.

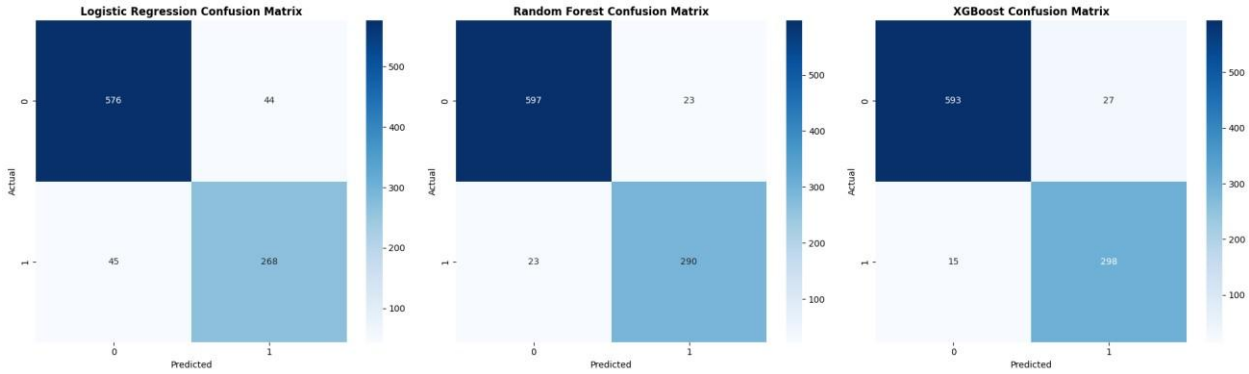

Fig 10: Confusion Matrix

XGBoost has the fewest number of false negatives and overall misclassification. That fits its high recall score and indicates that it is indeed excellent at determining the positive highquality wine samples precisely what the classification task requires.

10.3 Feature Importance and Model Interpretability

The importance of features in the three models was similar, and it is not a random coincidence of a single algorithm. In logistic regression, the largest weights occurred with alcohol, volatile acidity and sulphates, it is with that where the greatest signal is being produced. Alcohol raises the quality anticipated, whereas volatile acidity lowers it.

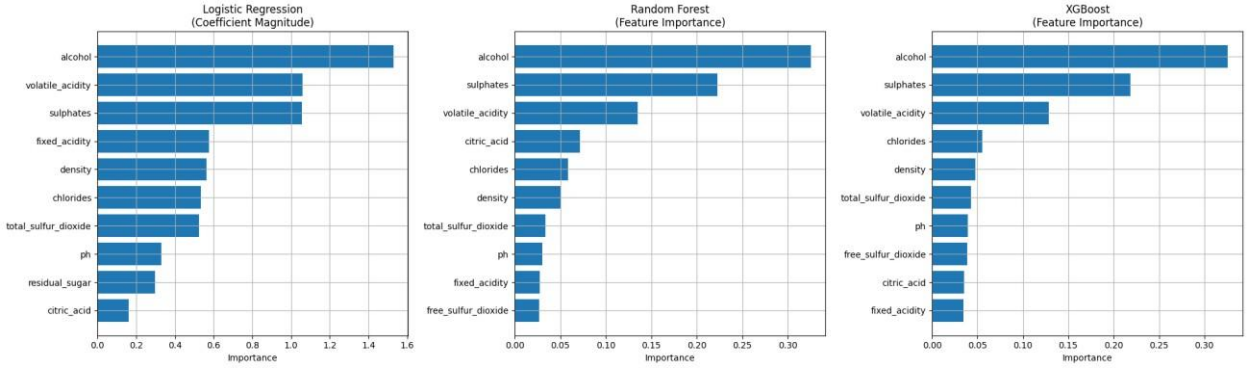

Fig 11: Feature Importance in different models

Both Random Forest and XGBoost algorithm ranks alcohol as the best feature then sulphates then volatile acidity as a rank. Other factors such as density, chlorides, and sulfur dioxide are loitering around the middle with some moderate impact. The similarity of all the three models makes us more confident that the associations are observed as valid and are in line with those which are already known by the literature on wine production.

10.4 Shapely Additive exPlanations (SHAP)

SHAP analysis reveals architecture of behavioral differences of such models. As an example, in the case of Logistic Regression, alcohol is very important, and tree-based models evenly distribute their feature significance. Random Forest raises sulphates, whereas XGBoost does exactly that by ranking it first place, thus relegating alcohol to a lower position in the list. Because XGBoost uses a more diverse range of features such as sulfur dioxides, it can capture multi-dimensional, non-linear interactions and that is why its predictive capability and strength is better than the base models.

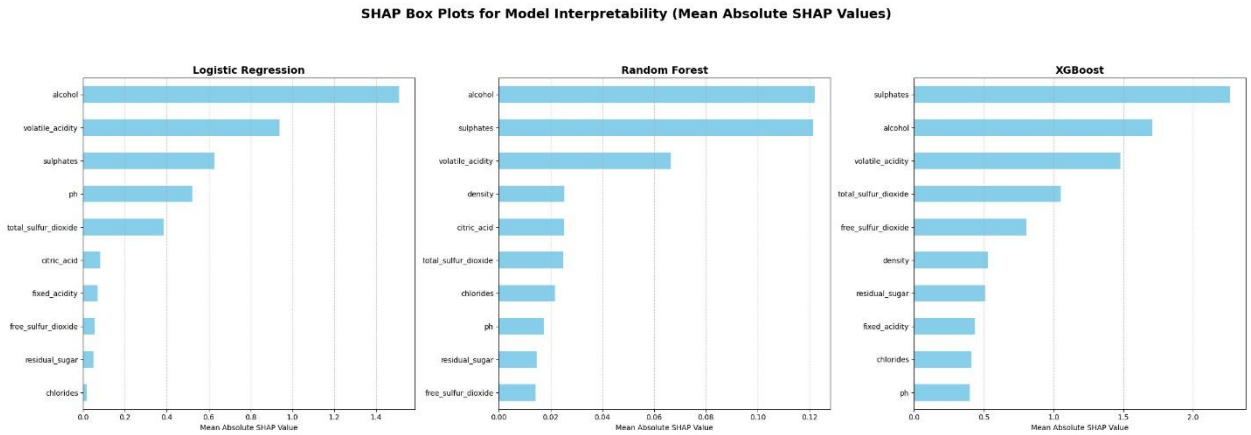

Fig 12: SHAP Boxplot

10.5 Learning Curve Analysis

Different model trends can be seen in the learning curves. In the case of Logistic Regression, there is high bias, converging loss, but still low accuracy, and is thus classified as underfitting. Although the ensemble results of the Random Forest are satisfactory, it possesses overfitting the smallest ensemble: this is due to the high generalization gap and slight overfitting. In contrast, of all the learning models, the XGBoost has the most success: it improves continuously in validation accuracy with the addition of more data and has low margins reflecting that the learner has improved generalization and can benefit from extra samples for the prediction of the wine quality.

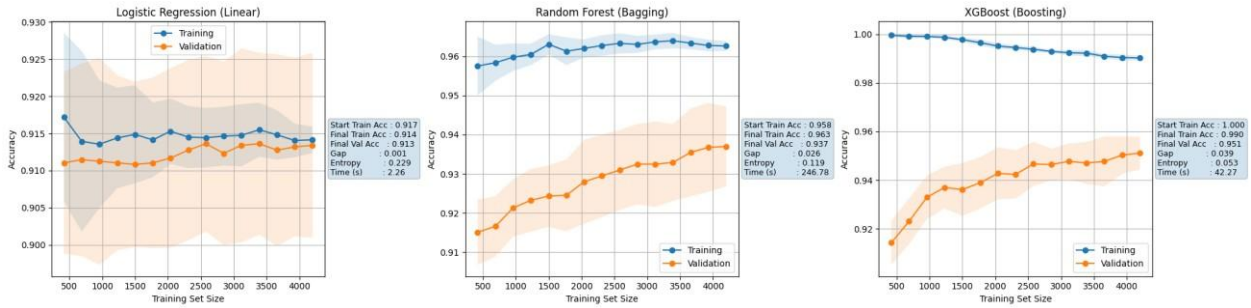

Fig 13: Training Validation Curve

10.6 Comparative Summary

Generally, the evaluation outcomes demonstrate that ensemble approaches are conclusively better than the basic linear baseline. Certainly, logistic regression is low cost, and easy to understand and apply, but it just does not capture the complexity. Random Forest is robust and solid, whereas XGBoost is the optimal being in having good performance, recall and being fast. All the metrics, confusion matrix, feature importance, graphs and everything are significant proof in favor of selecting XGBoost as the most optimal model in this project.

11 Conclusion

11.1 Summary

Thus, this project was the application of supervised machine learning when classifying the quality of wine based on physicochemical characteristics. Having completed all that exploratory data analysis, data cleaning, preprocessing, and used three models; Logistic Regression, Random Forest, and XGBoost, and then tested them within a stable experimental framework. The ensemble-based methods were significantly superior to the linear baseline; XGBoost gave the best accuracy, recall, F1-score, and AUC scores. The models were bought to have high generalization as confirmed by learning curve analysis and feature importance and SHAP based explainability revealed that alcohol content, sulphates, and volatile acidity were the most important predictors of wine quality.

11.2 Reflection on Individual Learning

The project provided a practical understanding of the machine learning pipeline, including data preparation and management class imbalance as well as model optimization and assessment. SHAP implementation enhanced understanding of model interpretability since the predictions are related to the contribution of each feature and this indeed highlighted the importance of transparency in predictive systems. The ability to explain model behavior in ways that can explain using explainable AI techniques helped to develop analytical skills and become more confident in the responsible implementation of such models.

11.3 Future Recommendations

The explainability analysis could be further expanded in the future by integrating instance level SHAP visualizations to facilitate decision making. Other enhancements might involve trying other imbalance handling methods like resampling and testing deep learning architectures. The inclusion of domain specific knowledge and addition of more features can also enhance predictive accuracy and model credibility.

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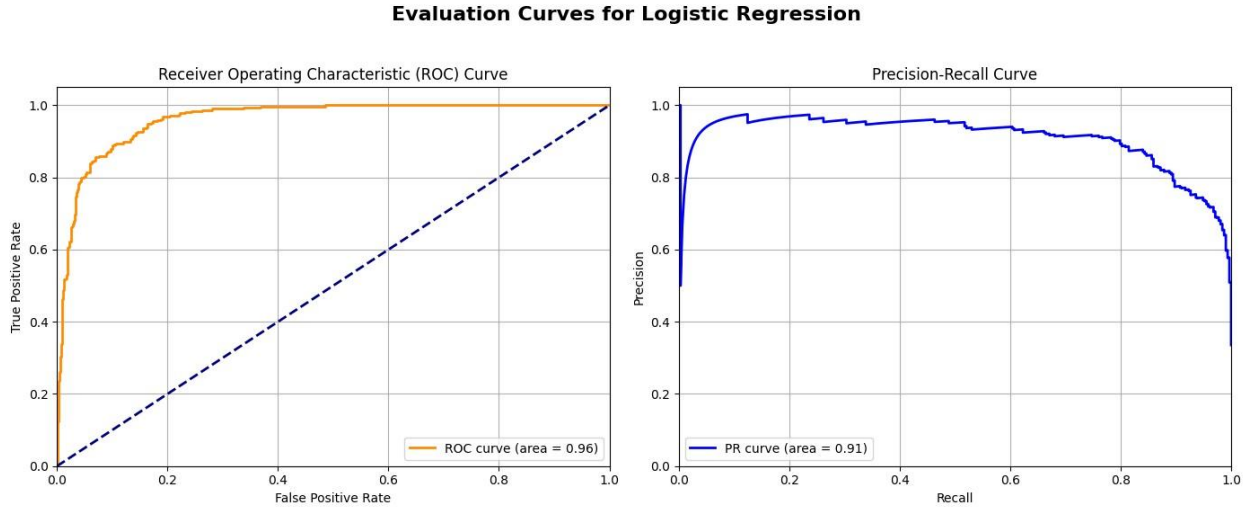
13 Appendix

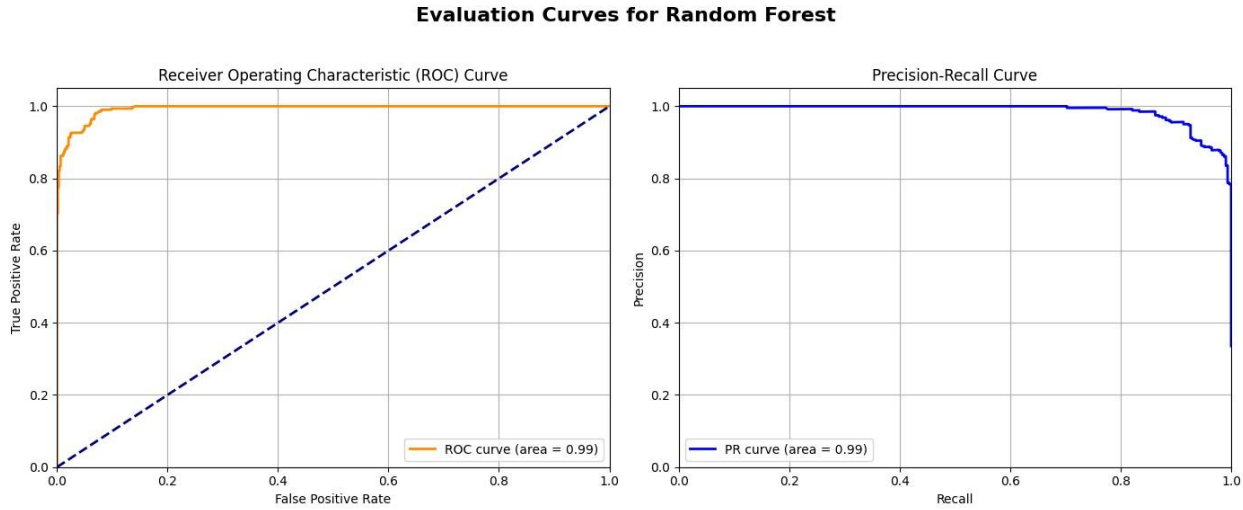
13.1 Appendix A: CODE

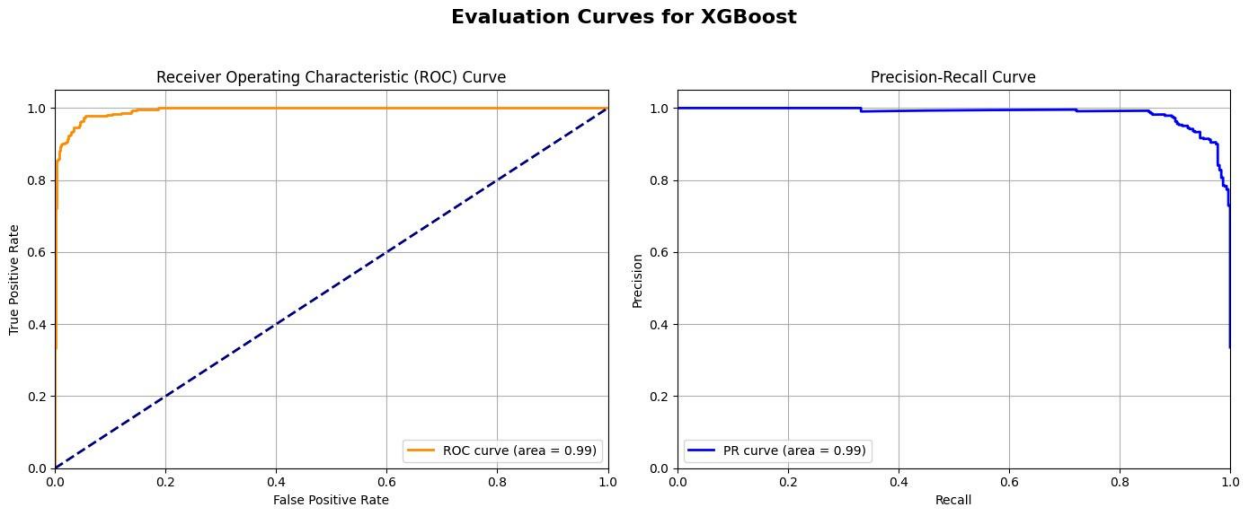
The file below is the google colab file used to train the wine quality:

https://colab.research.google.com/drive/1KCcAIuYMBwvUATOMZIPVxh2uNp5RJsB5?usp=sharing

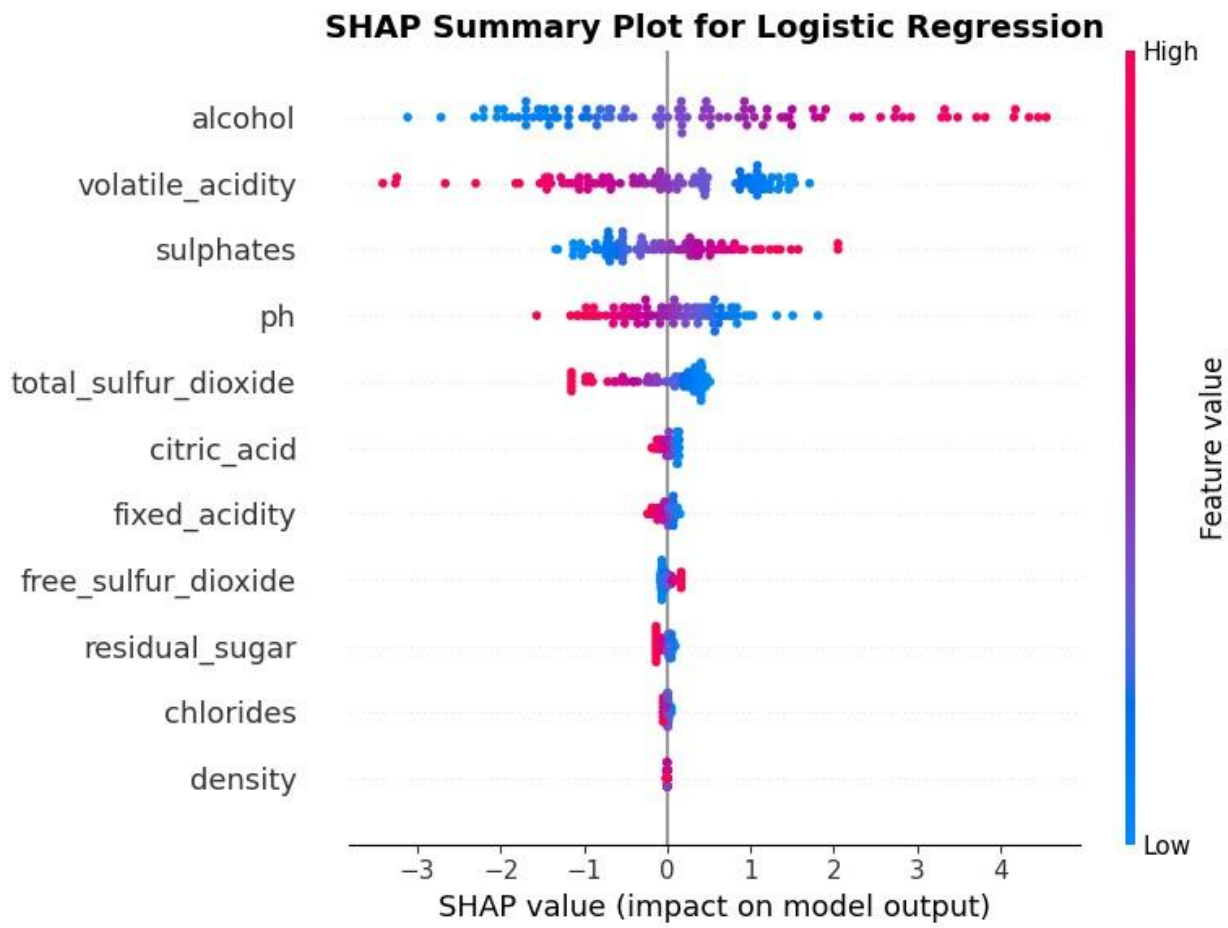
13.2 Appendix B: ROC and precision-recall curve

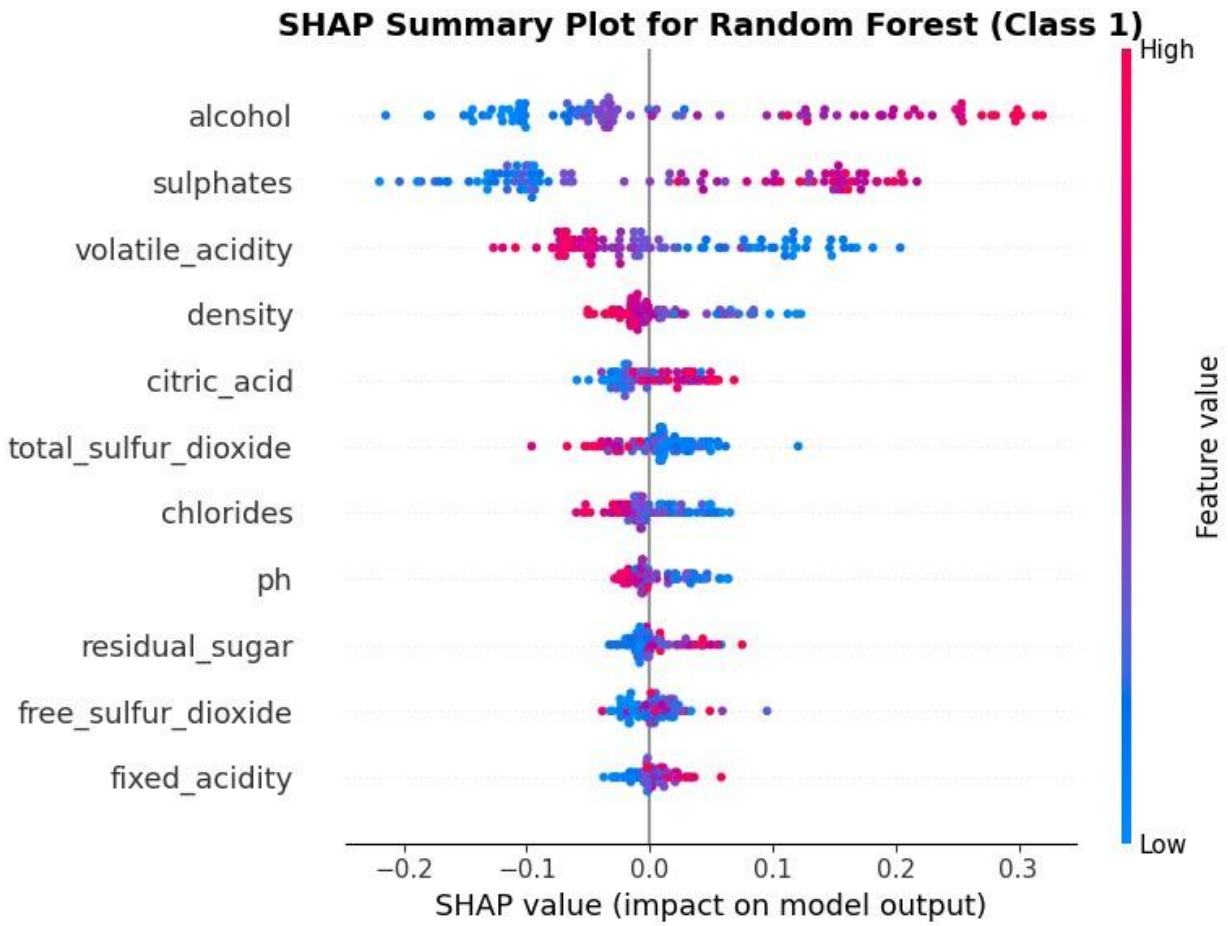


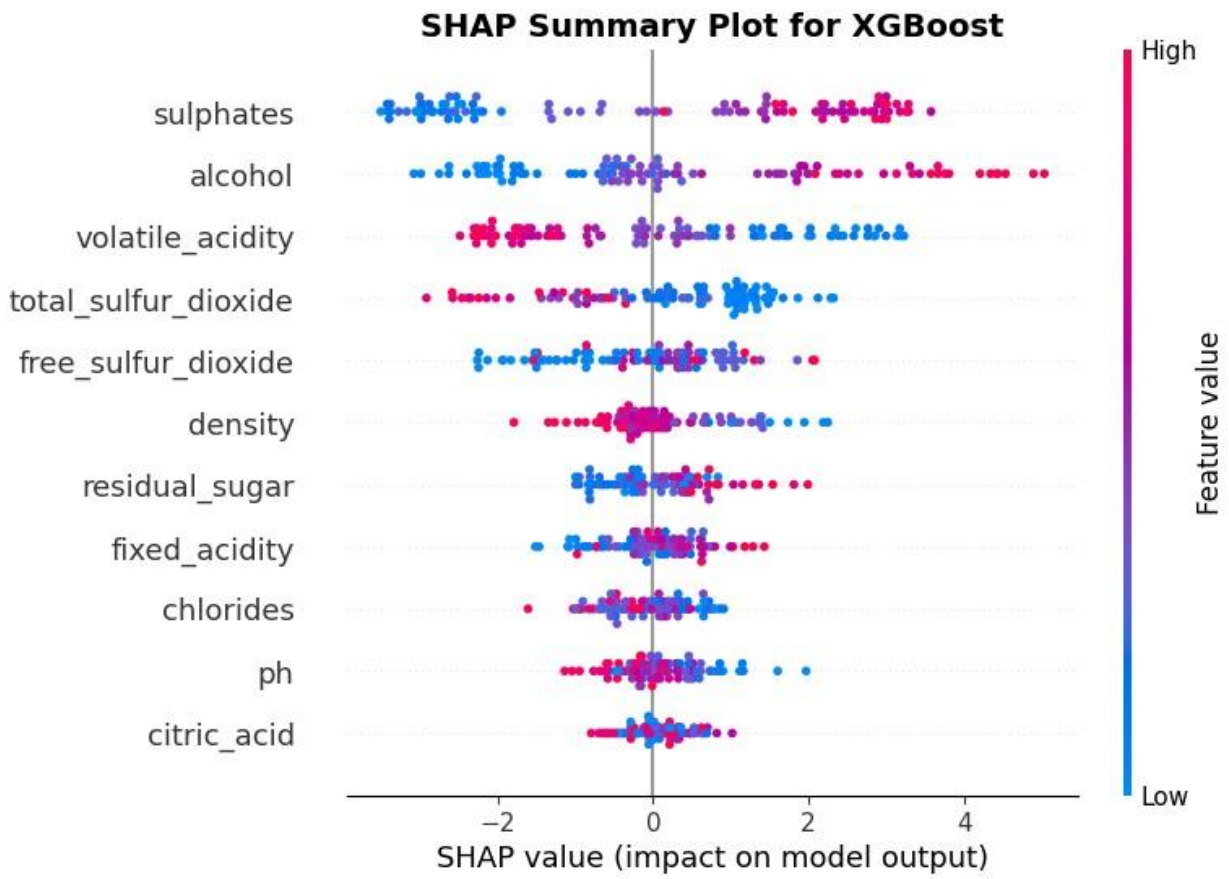




13.3 Appendix C: SHAP Summary Plot




SHAP Summary Plot for Random Forest (Class 1)
alcohol
sulphates
volatile_acidity
density
citric_acid
total_sulfur_dioxide
chlorides
ph
residual_sugar
free_sulfur_dioxide
fixed_acidity
−0.2
−0.1
0.0
0.1
0.2
0.3
SHAP value (impact on model output)
High
Feature value
Low


SHAP Summary Plot for XGBoost
sulphates
alcohol
volatile_acidity
total_sulfur_dioxide
free_sulfur_dioxide
density
residual_sugar
fixed_acidity
chlorides
ph
citric_acid
−2
0
2
4
SHAP value (impact on model output)
High
Feature value
Low